From Relation to Node:

Minimal Conditions for the Emergence of Grounded Symbolic Graphs,
with a Machine-Verified Formalization in Lean 4

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1 Motivation and Background

1.1 The originary hypothesis

Eric Gans's Generative Anthropology proposes that language, and with it the distinctively human forms of culture, originated in a single kind of event: the *originary scene* [2, 3]. In Gans's account, a group of hominids converge, through mimetic desire, on a single desirable object. Mutual imitation intensifies their rivalry beyond what an animal dominance hierarchy can contain, and the group faces a violent crisis. The crisis is deferred when the gesture of appropriation is aborted. The aborted gesture, shared and reciprocally understood, becomes the first sign: an *ostensive* that designates the object as a centre which no one may take. The scene concludes in the *sparagmos*, the collective division of the object.

From the ostensive, Gans derives the further elementary forms of language in sequence. The *imperative* arises from an ostensive made in the absence of its referent, which the hearer answers by supplying the missing object. The *declarative* arises when the imperative cannot be fulfilled, and the hearer preserves the relation symbolically instead. Later work in the tradition, notably by Adam Katz, develops related notions, among them the *centre* toward which the attention of a scene converges, and the *firstness* of the participant who defers first.

The ambition of the theory is captured in its name. It is *generative*: it claims to derive complex linguistic and cultural forms from a minimal beginning. Gans presents the originary hypothesis as a minimal one, positing a single event and asking what must follow from it [3, 5].

1.2 The problem: narrative content of uncertain necessity

This work grows out of a long engagement with Gans's writings and those of Adam Katz, and out of a persistent dissatisfaction with one feature of the account. The originary scene is not only a logical structure. It is also a story, and the story has specific content: mimetic desire as the source of convergence on the object, mimetic rivalry as the source of crisis, the prospect of catastrophic violence as what compels deferral, a sacred centre, and a sparagmos that resolves the scene by division. This content is inherited in part from René Girard's mimetic theory [4]. Each element is anthropologically suggestive, and each ties the origin of language to themes the theory develops elsewhere, such as the sacred, ethics and exchange.

It is not evident, however, which of these elements the derivation of the sign and its forms actually requires. Several are open to objection on their own terms. Violence among humans and other animals is common and rarely catastrophic; disputes end through exhaustion or

through hierarchy without producing signs. Refraining from an object one wants is, on ordinary assumptions about preference, a losing strategy, so a deferral that is supposed to be motivated by self-interest needs further explanation. And a collective division of the object adds a stage whose necessity for the emergence of reference is unclear. Much of the narrative therefore appears arbitrary relative to what it is meant to explain.

The issue is not whether these elements are plausible. It is whether they are *load-bearing*. If the derivation depends on them, it inherits their contingency and takes on the character of a just-so story. If it does not, it should be possible to state the derivation without them, and doing so would show what the generative core of the theory actually is.

1.3 Aim: minimal conditions for generativity

The present work attempts to separate that core from its narrative content. Its aim is to identify a minimal set of assumptions under which it can be *proved* that symbolic structure arises from conditions in which it is absent, and to make every assumption explicit. Generativity is given three precise senses:

Existence.
There is a model of the assumptions in which the structure arises from a prior state that lacks it.

Necessity.
In every model of the assumptions, whenever the structure exists, it has a determinate origin and order.

Universality.
The operations that produce the first symbolic structures suffice to produce any finite structure of the relevant kind.

The aim is not to show that such emergence was inevitable, nor that it happened in one particular way. It is to show that emergence is possible under weak conditions, and that its form is constrained by those conditions.

1.4 Why graphs

The structure whose emergence is at issue is taken to be relational structure in general, in the form of graphs. Any finite relational system can be represented as a graph. Relations among more than two terms can be encoded by an incidence construction, in which each relation instance becomes a node joined to its arguments. The question of how language begins is therefore approached structurally. Consider a world that contains physical objects and physical relations among them, but no symbolic nodes. How can symbolic nodes, and relations among them, arise in such a world? Physical objects are assumed. Symbolic nodes are not.

This framing makes precise a specific hypothesis that motivated the work: that

- the ostensive establishes a contextual relation, an edge;
- the imperative attempts to complete an unsatisfied relation;
- the declarative yields an independently representable symbolic object, a node.

One outcome of the formalization is a sharper version of this hypothesis. The first symbolic node is best understood as a *reified relation*: a relation that has itself become something one can refer to. Relations between nodes, in turn, arise by the same operation applied one level up.

1.5 Rival orderings

A further motivation is comparative. The sequence ostensive, imperative, declarative is one reading among several. Other interpretations, for instance accounts that give priority to the declarative, plausibly license themselves within their own axioms: each ordering may be derivable once suitable assumptions are made. If so, the substantive question is not which ordering is correct in the abstract. It is which assumptions each ordering requires, and how plausible those assumptions are. A framework in which every assumption is explicit allows such alternatives to be compared on equal terms. The present work develops one such framework and records, for each result, the assumptions on which it depends.

1.6 Why machine verification

Arguments in humanistic theory often rely on steps that are intuitively compelling but unstated. Formalization in a proof assistant forces every assumption to be written down and every inference to be checked. All results reported here are formalized in the Lean 4 proof assistant [1] and checked in full, with no unproven steps.

The process was itself informative. Several assumptions that the informal narrative leaves implicit were exposed only when a proof failed without them. Among them:

- a kinematic principle linking a reproduced gesture to the object it was originally directed at;
- the requirement that one participant defer before another, which corresponds to Katz's firstness;
- the dependence of falsehood on the drift of signs away from their iconic origins;
- the reliance of affirmation, derived as the negation of a negation, on classical logic.

Formalization also makes it possible to say exactly which statements are derived and which are merely definitions. The paper marks this distinction throughout.

1.7 Outline

Section 2 introduces the primitives and axioms, which concern only bodies, movements, reach and resemblance. Section 3 defines taking-as, the centre and the elementary modes as patterns of response. Section 4 reports the structural results that hold in every model. Section 5 reports the existence and universality results. Section 6 discusses the results, their relation to Generative Anthropology, and directions for further work.

References

[1] L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, Lecture Notes in Computer Science 12699, pages 625–635. Springer, 2021.

[2] E. Gans. *The Origin of Language: A Formal Theory of Representation.* University of California Press, Berkeley, 1981.

[3] E. Gans. *Originary Thinking: Elements of Generative Anthropology.* Stanford University Press, Stanford, 1993.

[4] R. Girard. *Violence and the Sacred*. Translated by P. Gregory. Johns Hopkins University Press, Baltimore, 1977.

[5] A. Katz, editor. *The Originary Hypothesis: A Minimal Proposal for Humanistic Inquiry*. The Davies Group, Aurora, CO, 2007.